Gabriel Cavalcanti Alves

Game Developer | Software Engineer

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Professional Summary

I am a Junior Game Developer and Software Engineer with hands-on experience in **Unity 3D/2D** and **full-stack web development**. Proven ability to design and implement gameplay systems, optimize performance, and collaborate in agile teams. Passionate about creating engaging, player-focused experiences and scalable software solutions.

Key Skills

- **Programming & Scripting: C#**, Java, JavaScript, TypeScript, C++, C, HTML5, CSS3
- **Game Development: Unity (2D & 3D)**, **Gameplay Programming**, UI/UX Systems, Animation, Physics, AI, Level Design
- **Web & Backend:** Vue.js, Vite, Vuetify, Spring Boot, Node.js, Strapi, REST APIs, PostgreSQL, MySQL, DBeaver
- **Mobile & Tools:** Android Studio, Firebase, Git, Git Bash, GitLab, Figma, Adobe XD, Astah
- **Methodologies & Practices: Agile/Scrum**, **OOP**, DevOps Practices, **CI/CD**, UI/UX Design, Automated Testing (Playwright)
- **Languages: English (**Professional Proficiency **– B2/C1)**, Portuguese (Native)

Projects

Cook & Conquer – 2D Roguelike Game | Unity, C#

- Helped a kickstart team to develop a complete 2D roguelike game.
- Created an inventory system, drop system, collect system and crafting system.
- Designed UI for listing inventory items, a crafting item area and item slots for usage.

Astro Adventure 3D – 3D Space Survival Game | Unity, C#

- Developed complete 3D gameplay systems including character controllers, environmental interactions, and survival mechanics.
- Integrated animations, sound effects, and particle systems to create an immersive alien world.
- Designed UI for health, inventory, and mission tracking.

Astro Obstacle – 3D Infinite Runner | Unity, C#

- Created procedurally generated obstacle courses with increasing difficulty curves.
- Implemented visual and auditory feedback systems to enhance player engagement.

Lost in The Stars – 2D Action-Platformer | Unity, C#

- Built 2D movement systems, interactive level design, and ability-based progression.
- Implemented discovery-driven gameplay where player abilities unlock new areas and challenges.

MyPong – Multiplayer Pong Game | Unity, C#

- Modernized classic Pong with customizable player colors, controls, and local multiplayer support.
- Designed an intuitive UI for seamless match setup and gameplay.

4Kingdoms – 4-Player Local Multiplayer Brawler | Unity, C#

- Developed combat systems, character abilities, and responsive controls for local multiplayer sessions.
- Designed intuitive UI for player selection, match setup, and in-game feedback.

Professional Experience

Software Engineer Intern (*Aug 2024 – Jul 2025*)

Center for Informatics Practices (NPI) – Federal University of Ceará | Quixadá, Brazil

- Developed a full-stack nursing home management system using Vue.js, Vuetify, TypeScript, Spring Boot, and PostgreSQL.
- Engineered responsive interfaces, implemented form validations, performed CSS/UI improvements across multiple modules and optimized front-end performance.
- Built and maintained RESTful APIs, managed database relationships, and resolved data integrity issues using Spring Boot and DBeaver.
- Conducted automated testing with Playwright, resolved HTTP 500 errors and fixed critical bugs, improving system stability and data validation.
- Enhanced system performance through pagination, sorting functionalities, and table responsiveness.
- Refactored code by improving data typing (varchar → TIME) and replacing inline styles with Vuetify utility classes.
- Collaborated in Scrum-based teams, using GitLab, SonarQube and CI/CD pipelines for version control and quality assurance.
- Used Figma and GitLab for task management, UI review, and team collaboration (Miro for spring review).

Education

Bachelor of Software Engineering (*Feb 2020 – Aug 2025*)

Federal University of Ceará | Quixadá, CE, Brazil

- **Relevant Coursework:** Object-Oriented Programming, Database Systems, Agile Methodologies, UI/UX Design, Web Development, Mobile Development, Game Development.

Certifications

- **Profession: Game Designer and Unity Developer** – **EBAC (2024–2025)**
- **Scrum Foundation Professional Certificate (SFPC)** – **CertiProf**

Published Research: "The Players' Perception Over The Artificial Intelligence in Digital Games: An Exploratory Study With **Final Fantasy XV" – SBGames 2025 (**DOI**)**